Solution 1:

(a) **Assumptions:**

- **QDA:** Gaussian class-conditionals $\mathbf{x}|y = k \sim \mathcal{N}(\boldsymbol{\mu_k}, \Sigma_k)$, where each class has its own mean vector $\boldsymbol{\mu_k}$ and covariance matrix Σ_k; no feature independence assumed. As the Σ_k can differ between classes, decision boundaries are quadratic.
- **LDA:** special case of QDA with class-specific $\boldsymbol{\mu_k}$ but shared covariance matrix $\Sigma_k = \Sigma$ for all k, resulting in linear decision boundaries.
- **NB:** conditional independence of the features given the class:

$$p(\mathbf{x}|y = k) = \prod_{j=1}^{p} p(x_j|y = k)$$

With a univariate Gaussian per feature (Gaussian NB), this is a special case of QDA with diagonal covariance matrices $\Sigma_k = \text{diag}(\sigma_{k1}^2, \ldots, \sigma_{kp}^2)$: variances may differ, all covariances are zero.

(b) **Assignment:** A is QDA (most general, class-specific Σ_k); B and C are LDA and Gaussian NB (either way round), both special cases of A:

- A $\rightarrow$ LDA: shared $\Sigma_k = \Sigma$ for all k.
- A $\rightarrow$ NB: diagonal $\Sigma_k = \text{diag}(\sigma_{k1}^2, \ldots, \sigma_{kp}^2)$.
- Intersection of LDA and NB: shared **and** diagonal, $\Sigma_k = \Sigma = \text{diag}(\sigma_1^2, \ldots, \sigma_p^2)$ for all k.

(c) **Specific assumption:** the class-conditional distributions are Gaussian. NB can also be used with other (non-Gaussian) feature distributions; then NB is no longer a subset of QDA and its region would partly lie outside of A.

Solution 2:

All methods estimate $K - 1$ class priors π_k (as $\sum_{k=1}^{K} \pi_k = 1$) and K class means $\boldsymbol{\mu_k} \in \mathbb{R}^p$, i.e., Kp parameters. A symmetric $p \times p$ covariance matrix has p variances and $(p^2 - p)/2$ covariances, i.e., $p + (p^2 - p)/2 = p(p + 1)/2$ parameters.

(a) **LDA**, $\mathbf{x}|y = k \sim \mathcal{N}(\boldsymbol{\mu_k}, \Sigma)$ with one covariance matrix shared by all classes:

$$\underbrace{(K-1)}_{\pi_k} + \underbrace{Kp}_{\boldsymbol{\mu_k}} + \underbrace{\frac{p(p+1)}{2}}_{\Sigma}$$

(b) **QDA**, $\mathbf{x}|y = k \sim \mathcal{N}(\boldsymbol{\mu_k}, \Sigma_k)$ with one covariance matrix per class:

$$\underbrace{(K-1)}_{\pi_k} + \underbrace{Kp}_{\boldsymbol{\mu_k}} + \underbrace{K \cdot \frac{p(p+1)}{2}}_{\Sigma_k}$$

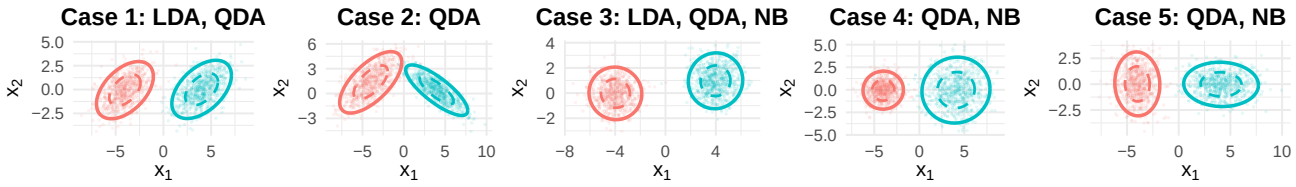
(c) **Gaussian NB**, features conditionally independent given the class, $x_j|y = k \sim \mathcal{N}(\mu_{kj}, \sigma_{kj}^2)$ for $j = 1, \ldots, p$, i.e., diagonal $\Sigma_k = \text{diag}(\sigma_{k1}^2, \ldots, \sigma_{kp}^2)$:

$$\underbrace{(K-1)}_{\pi_k} + \underbrace{Kp}_{\mu_{kj}} + \underbrace{Kp}_{\sigma_{kj}^2} = (K-1) + 2Kp$$

For comparison: (multinomial) logistic regression has $(K - 1)(p + 1)$ parameters, i.e., $p + 1$ for $K = 2$.

Solution 3: Extra task

The plot titles show which classifiers can model each case; the table gives the reasons.



Case	Σ_1, Σ_2	Can model	Cannot	Why
1	$\Sigma_1 = \Sigma_2 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$	LDA, QDA	NB	Shared but non-diagonal Σ (correlation): linear boundary; NB cannot model correlation.
2	$\begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}, \begin{pmatrix} 3 & -2 \\ -2 & 2 \end{pmatrix}$	QDA	LDA, NB	Different full Σ_k (different correlation): quadratic boundary with cross-term $x_1 x_2$; only QDA allows class-specific full Σ_k.
3	$\Sigma_1 = \Sigma_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	LDA, QDA, NB	–	Shared and spherical $\Sigma_k = \Sigma = \boldsymbol{I}$: all assumptions hold, linear boundary.
4	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$	QDA, NB	LDA	Spherical but different scales ($\Sigma_2 = 3\Sigma_1$): axis-aligned quadratic boundary (a circle); LDA needs a shared Σ.
5	$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$	QDA, NB	LDA	Diagonal (no correlation) but elongated along different axes: axis-aligned quadratic boundary; LDA needs a shared Σ.